Technical Guide: Enabling High-Speed 5 GHz Hotspots on Linux

Overcoming Regulatory Locks and Legacy Fallbacks

Documentation

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5 GHz Hotspot Optimization

This guide details the configuration required to successfully launch a 5 GHz Wi-Fi hotspot on Linux systems that default to restrictive regulatory domains. It addresses critical issues where the 5 GHz Access Point (AP) fails to start entirely or operates at severely reduced speeds.

1. Problem Overview

Prior to applying this configuration, the system may exhibit the following behavior:

- **Complete Failure of 5 GHz AP:** The hotspot software cannot initiate a 5 GHz network, forcing the system to rely solely on the crowded and slower 2.4 GHz band.
- **Regulatory "No-IR" Lock:** The wireless card's firmware flags most 5

GHz channels as “No Initiate Radiation” (No-IR), preventing it from acting as a master (AP).

- **Speed Limitations:** If an AP is forced, it may fallback to legacy 802.11a protocols (max 54 Mbps), failing to utilize modern 802.11ac (Wi-Fi 5) capabilities.

2. Hardware and Capability Verification

Before configuration, use the following commands to identify your hardware and verify its supported frequencies:

- **Identify Interface:** Use `iw dev` to find your wireless interface name (e.g., `wlan0`).
- **Check Hardware Capabilities:** Use `iw phy` (or `iw phy0 info`) to view detailed capabilities, supported bands, and whether a channel is marked as `(no IR)`.

3. Implementation

The following configuration should be applied to `/etc/create_ap.conf`. This setup lifts “World” (00) domain restrictions and forces the use of Wi-Fi 5 performance features.

Optimized Configuration (`/etc/create_ap.conf`)

```
CHANNEL=149
GATEWAY=192.168.12.1
WPA_VERSION=2
ETC_HOSTS=0
DHCP_DNS=gateway
NO_DNS=0
NO_DNSMASQ=0
HIDDEN=0
MAC_FILTER=0
MAC_FILTER_ACCEPT=/etc/hostapd/hostapd.accept
ISOLATE_CLIENTS=0
SHARE_METHOD=nat
IEEE80211N=1
IEEE80211AC=1
IEEE80211AX=0
HT_CAPAB=[HT40+]
VHT_CAPAB=[MAX-MPDU-3895][SHORT-GI-80][TX-STBC-2BY1]
DRIVER=nl80211
NO_VIRT=0
COUNTRY=US
FREQ_BAND=5
NEW_MACADDR=
DAEMONIZE=0
DAEMON_PIDFILE=
DAEMON_LOGFILE=/dev/null
DNS_LOGFILE=
NO_HAVEGED=0
WIFI_IFACE=wlan0
```

```
INTERNET_IFACE=eth0
SSID=YOUR_SSID
PASSPHRASE=1234567890
USE_PSK=0
ADDN_HOSTS=
```

4. Launch Methods

The hotspot can be started using the configuration file via the terminal or through the graphical interface.

Option A: Command Line (CLI)

Start the access point manually using the optimized config:

```
sudo create_ap --config /etc/create_ap.conf
```

Option B: Graphical User Interface (GUI)

You can use the **Linux WiFi Hotspot** application. Launch it from your desktop environment or via terminal:

```
wihotspot
```

5. Advanced: Running Multiple Hotspots

It is possible to run two separate access points simultaneously (e.g., one 5 GHz on `wlan0` and one 2.4 GHz on `wlan1`) while sharing the same internet connection (`eth0`).

Critical Rule: Each instance must use a **unique Gateway IP subnet**. If both use the default `192.168.12.1`, IP conflicts will prevent them from working.

Setup Instructions

1. **Interface 1 (`wlan0` - 5 GHz):** Use the standard `/etc/create_ap.conf` with: `ini      WIFI_IFACE=wlan0      GATEWAY=192.168.12.1`

2. **Interface 2 (`wlan1` - 2.4 GHz):** Create a new config `/etc/create_ap_wlan1.conf` with a different gateway: `ini` `WIFI_IFACE=wlan1      GATEWAY=192.168.13.1   <-- Must be unique!` `SSID=MySecondHotspot      CHANNEL=6`

Execution

Launch both instances as background daemons:

```
sudo create_ap --config /etc/create_ap.conf --daemon
sudo create_ap --config /etc/create_ap_wlan1.conf --daemon
```

6. Technical Parameter Breakdown

`COUNTRY=US` (Regulatory Unlock)

Lifts the "No-IR" flag from upper 5 GHz channels (149-161). **Without this, the 5 GHz AP will likely fail to start.**

`VHT_CAPAB` (Performance Unlock)

- **`[MAX-MPDU-3895]`:** Increases frame efficiency.
- **`[SHORT-GI-80]`:** Boosts throughput by ~10% via 400ns Guard Intervals.
- **`[TX-STBC-2BY1]`:** Improves signal reliability.

7. Verification

Once a client is connected, verify the transmission speed:

```
sudo iw dev ap0 station dump
```

Look for the `tx bitrate`. A successful configuration should show speeds significantly higher than 54 MBit/s (e.g., **433.3 MBit/s** or **866.7 MBit/s**).